Malihe Aliasgari, Ph.D.

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School of Integrative Science and Technology, Kean University Union, New Jersey, USA

Research Areas

- Artificial intelligence and machine learning; large language models and recommender systems; information theory, privacy, and distributed computing; coding theory; lattices, Gröbner bases, and algebraic decoding.

Academic Appointments

2023 – Present — **Assistant Professor**, Department of Mathematical Sciences, Kean University, Union, NJ.
2021 – 2023 — **Postdoctoral Researcher**, School of Integrative Science and Technology, Kean University, Union, NJ.
2020 – 2021 — **Adjunct Professor**, School of Integrative Science and Technology, Kean University, Union, NJ.
Jan.–Jul. 2014 — **Visiting Researcher**, Carleton University, Ottawa, Canada.
2013 – 2016 — **Assistant Professor**, Islamic Azad University, Roudehen Branch, Tehran, Iran.

Education

2016 – 2020 — **Ph.D., Electrical Engineering**, New Jersey Institute of Technology (NJIT), GPA: 4.0/4.0.
Dissertation: *Coding Against Stragglers in Distributed Computation Scenarios*.

2010 – 2014 — **Ph.D., Pure Mathematics**, Amirkabir University of Technology, Iran, GPA: 4.0/4.0.
Dissertation: *Decoding Lattices via Gröbner Bases*.

2006 – 2008 — **M.Sc., Pure Mathematics**, Amirkabir University of Technology, Iran, GPA: 4.0/4.0.
Thesis: *Persistent Homology and Localized Homology*.

2002 – 2006 — **B.Sc., Pure Mathematics**, Amirkabir University of Technology, Iran, GPA: 3.8/4.0.

Publications

Manuscripts Under Review / Submitted

- *Encoding-Sensitive Nullstellensatz Certificates for Bounded-Support Syndrome Exclusion in LDPC Codes*, M. Aliasgari, H. Khodaiemehr, and K. Bagheri. Submitted to *IEEE Information Theory Workshop (ITW) 2027*.
- *Reimagining the Newsvendor Game: A Generative AI Approach to Inventory Decision*, S. Shokoohyar, M. Aliasgari, and Y. Nejatbakhsh. Submitted to *Decision Sciences Journal of Innovative Education*, 2026.
- *Query-Formulation Sensitivity in Citizen-Facing Public-Health RAG: A Five-Retriever Retrieval Comparison and a Single-Configuration End-to-End Study*, M. Aliasgari, Y. Nejatbakhsh, and P. Safaee. Under review at *Discover Artificial Intelligence*, September 2026.
- *A Safety-Constrained Hybrid Food Recommender System for Phenylketonuria: Integrating Cuisine-Aware Filtering and Clinical Rule Enforcement*, M. Aliasgari, M. Ajetunmobi, Y. Nejatbakhsh, and R. Nejatbakhsh. Under review at *JMIR Formative Research*, April 2026.
- *Low-Unsatisfaction Screening for QC-LDPC Code Selection in Wireless Links*, M. Aliasgari. Manuscript under revision following review at *IEEE Wireless Communications Letters*, 2026.
- *Functional Converses for Private Matrix-Product Retrieval under Library-Wide Decoder-Side Correlation*, M. Aliasgari and Y. Nejatbakhsh. Submitted to *IEEE Transactions on Information Theory*, August 2026.

Peer-Reviewed Journal Articles

- *Routing Leakage in Adaptive Coded Distributed Matrix Multiplication: An Information-Theoretic Framework for Conditional Differential Privacy*, M. Aliasgari. *IEEE Access*, vol. 14, pp. 100918–100934, 2026.
- *Enhancing Stock Market Prediction with Hybrid Deep Learning: Integrating LSTM, Transformer Attention, Federated Learning, and Sentiment Analysis*, Y. Nejatbakhsh and M. Aliasgari. *IEEE Access*, vol. 14, pp. 3926–3942, 2025.
- *Linear Label Code of a Root Lattice Using Gröbner Bases*, M. Aliasgari, D. Panario, and M.-R. Sadeghi. *Applicable Algebra in Engineering, Communication and Computing*, vol. 35, pp. 3–15, 2024.
- *Private and Secure Distributed Matrix Multiplication with Flexible Communication Load*, M. Aliasgari, O. Simeone, and J. Kliewer. *IEEE Transactions on Information Forensics and Security*, vol. 15, pp. 2722–2734, 2020.
- *Coded Computation Against Processing Delays for Virtualized Cloud-Based Channel Decoding*, M. Aliasgari, J. Kliewer, and O. Simeone. *IEEE Transactions on Communications*, vol. 67, pp. 28–38, 2019.

Publications (continued)

- *An Algebraic Method for Decoding q-ary Codes via Submodules of $\mathbb{Z}^n$*, M. Aliasgari and M.-R. Sadeghi. *IEEE Communications Letters*, vol. 18, no. 5, pp. 857–860, 2014.
- *Gröbner Bases for Lattices and an Algebraic Decoding Algorithm*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. *IEEE Transactions on Communications*, vol. 61, pp. 1222–1230, 2013.

Peer-Reviewed Conference Proceedings

- *The Voice That Tells in Different Languages: A Structured Review of Speech Biomarkers for Cognitive Decline in Ageing, MCI, and Dementia*, P. Safaee, K. Humlen, M. Aliasgari, Y. Nejatbakhsh, M. Mohammadi, L. Beijer, M. Gholami, and M. Noroozian. *Living Adaptations: Housing and Care Innovation*, Antwerp, Belgium, May 2026.
- *Recommending Healthy Food for Phenylketonuria with Collaborative and Content-Based Filtering*, M. Aliasgari, H. Li, Y. Zhang, Y. Nejatbakhsh, and R. Nejatbakhsh. *IEEE International Conference on E-health Networking, Application & Services (HealthCom)*, pp. 546–552, Japan, 2024.
- *Distributed Coded Computation*, M. Aliasgari. 25th Conference on Applications of Computer Algebra (ACA), Gebze–Istanbul, Turkey, 2022.
- *Linear Label Code of a Lattice Using Gröbner Bases*, M. Aliasgari, D. Panario, and M.-R. Sadeghi. 25th Conference on Applications of Computer Algebra (ACA), Gebze–Istanbul, Turkey, 2022.
- *Distributed and Private Coded Matrix Computation with Flexible Communication Load*, M. Aliasgari, O. Simeone, and J. Kliewer. *IEEE International Symposium on Information Theory (ISIT)*, pp. 1092–1096, 2019.
- *Coded Computation Against Straggling Decoders for Network Function Virtualization*, M. Aliasgari, J. Kliewer, and O. Simeone. *IEEE International Symposium on Information Theory (ISIT)*, pp. 711–715, 2018.
- *Johnson-Type Bound for Group Codes and Lattices*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. Conference on Applications of Computer Algebra (ACA), 2015.
- *Binomial Ideal Associated to a Lattice and Its Label Code*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. Conference on Applications of Computer Algebra (ACA), 2015.
- *Minimizing Exceptional Elements and Voids in a Cell Formation Problem with Optimal Number of Cells*, M. Aliasgari, M. Kazemi, R. Kia, and A. Ebrahimi. International Conference on Computers and Industrial Engineering (CIE45), Metz, France, pp. 1286–1294, 2015.
- *Gröbner Bases for Lattices and an Algebraic Decoding Algorithm*, M. Aliasgari, M.-R. Sadeghi, and D. Panario. Forty-Ninth Annual Allerton Conference on Communication, Control, and Computing, Illinois, USA, pp. 1414–1415, 2011.

Selected Research Presentations and Posters

2025
- *PKU-Recipe AI: Personalized PKU-Safe Recipe Generator*, IUSE/IST Research Day, poster presentation.
- *A PKU-Aware Hybrid Recommender System for Safe and Personalized Dietary Management*, IST Research Day, poster presentation.
- *A Smart Assistant for Phenylketonuria (PKU)*, IST Research Day, poster presentation.
- *PKU Food Checker: A Telegram-Based AI Assistant for PKU Dietary Monitoring*, IST Research Day, poster presentation.
- *PKU Scanner AI: A Mobile Tool for Safe Food Scanning*, IUSE/IST Research Day, poster presentation.
- *Using n8n to Create a Phenylketonuria AI Assistant*, IST Research Day, poster presentation.
- *A Hybrid Recommender System for Personalized Dietary Management in Phenylketonuria (PKU)*, Kean Research Day, poster presentation.
- *LLM-Powered Personalized Dietary Agent for Phenylketonuria*, Kean Research Day, poster presentation.
- *Virtual Avatar Development System*, Kean Research Day, poster presentation.
- *Automated Dementia Behavior Analysis and Empathetic Mapping Using Large Language Models*, Kean Research Day, poster presentation.

2024
- *LLM-Powered Personalized Dietary Agent for Phenylketonuria*, IUSE Scholars Academy, Kean University, poster presentation.
- *Linear Label Code of a Lattice*, Third Annual Kean University End-of-Year Faculty Research Conference.

2023
- *Stock Forecasting Using Advanced LSTM Models*, IST Research Day, Kean University, poster presentation.
- *The Power of Artificial Intelligence for Phenylketonuria*, IST Research Day, Kean University, poster presentation.
- *Comparing Predictive Models for Stock Market Analysis: Evaluating the Excellence of LSTM*, Kean Research Day, poster presentation.
- *Private Distributed Computation*, MAA New Jersey Section, Kean University, NJ.

2022
- *Private Distributed Coded Computation*, Applications of Computer Algebra (ACA), Turkey.
- *Linear Label Code of a Root Lattice Using Gröbner Bases*, Applications of Computer Algebra (ACA), Turkey.

2019
- *Distributed and Private Coded Matrix Computation with Flexible Communication Load*, North-American School of Information Theory, Boston, MA, poster presentation.

Publications (continued)

- *Distributed and Private Coded Matrix Computation with Flexible Communication Load*, IEEE ISIT, Paris, France.

2018
- *Coded Computation Against Processing Delays*, Center for Wireless Information Processing Graduate Student Seminar, NJIT.

2017
- *Coded Computation for Distributed Channel Decoding*, Center for Wireless Information Processing Graduate Student Seminar, NJIT.
- *Coded Computation for Distributed Channel Decoding*, North-American School of Information Theory, Atlanta, GA, poster presentation.

2016
- *Gröbner Bases for Lattices and Algebraic Decoding Algorithm*, Graduate Student Seminar, NJIT.

2013
- *On the Ideal Associated to a Lattice and Gröbner Bases of Some Root Lattices*, Groups, Group Rings, and Related Topics, United Arab Emirates University.
- *Maximum Likelihood Decoding for q-ary Codes via Submodules of $\mathbb{Z}_n$*, 44th Annual Iranian Mathematics Conference, Mashhad, Iran.
- *On the Binomial Ideal Associated to a Lattice and Its Label Code*, 44th Annual Iranian Mathematics Conference, Mashhad, Iran.

2012
- *An Algebraic Method for Solving the Closest Vector Problem in a Lattice*, 22nd Iranian Algebra Seminar, Hakim Sabzevari University, Iran, poster presentation.

2010
- *Decoding Group Codes with Gröbner Bases*, 42nd Annual Iranian Mathematics Conference, Rafsanjan, Iran, poster presentation.

Grant Activity

External Proposals

2026
- *Advancing Early Childhood PKU Care Through Parenting Education and AI*. Caplan Foundation for Early Childhood. **M. Aliasgari**, Y. Nejatbakhsh, and R. Singh. Amount requested: $50,000. Submitted January 2026.

2025
- *PRECISE: Personalized AI Recommendations for Empowering Care in Supporting Everyday Life Diet and Behavior in Adults with PKU*. NPKUA. **M. Aliasgari**, Y. Nejatbakhsh, and R. Singh. Originally submitted June 2024; not funded; resubmitted June 2025.

Proposal Consulting

- Consultant, NSF proposal *Length Scale Dependence of Extracellular Matrix Alignment in Tumor Growth and Invasion*. PI: Amir K. Miri, New Jersey Institute of Technology. Proposal amount: $600,000. Submitted 2025.

Awards, Fellowships, and Research Support

2026
- June Activities Award, Kean University, $2,000.
- Graduate Research Assistantship support, Kean University.

2025
- IST Fellowship, Kean University, $10,000.
- Participant, Kean University Shark Tank, *Smart Mind Care*.

2024
- Participant, Kean University Shark Tank, *AI for PKU*.
- Nominee, Faculty Research Mentor of the Year Award (Early Career Faculty), Kean University.

2023
- IST Fellowship, Kean University, $10,000.

2019
- Summer Research Assistantship Award, NJIT, $5,000.

2018–2019
- Research Assistantship Award, NJIT, $32,000.

2018
- Summer Research Assistantship Award, NJIT, $5,000.

2017–2018
- Research Assistantship Award, NJIT, $33,000.

2017
- Travel Award, NJIT, $600.

2016–2017
- Research Assistantship Award, NJIT, $52,000.

2015
- CIE45 Travel Award, Islamic Azad University, Roudehen Branch, $2,000.

2014
- NSERC Grant support, Carleton University, $12,000.

2013
- Sabbatical Award, Amirkabir University of Technology, $10,000.

Teaching Experience

Kean University

- Fall 2026 — Precalculus; Calculus I.
- Summer 2026 — Large Language Models.
- 2025 — Detection and Estimation Theory: Inference and Machine Learning for Engineers; Introduction to Deep Learning.
- Spring 2024 — Deep Learning and Recommender Systems.
- 2024–2025 — Discrete Mathematics.
- 2023–2024 — Research Federate Initiative (RFI).
- 2022–2023 — Special Topics: Machine Learning.
- 2022 — Information Theory and Coding.
- 2020–2024 — Mathematics/Computational Methods I; Precalculus; Mathematics/Computational Methods II; Calculus I; associated laboratories.

New Jersey Institute of Technology

- 2019 — Digital Communication Systems.

Earlier Teaching

- 2014–2015 — LaTeX and Beamer Workshops for faculty, Islamic Azad University, Tehran, Iran.
- 2013 — Engineering Mathematics and Ordinary Differential Equations, Islamic Azad University, Iran; Error Control Coding, Amirkabir University of Technology.
- 2009–2012 — Ordinary Differential Equations and Calculus II, Amirkabir University of Technology / Shahid Rajaee Teacher Training University / Islamic Azad University, Iran.
- 2009 — Statistics and Calculus I, Islamic Azad University, Iran.
- 2006–2008 — Mathematics Olympiad instruction, Iran.

Teaching Assistantships

- 2018 — Stochastic Signals and Systems, New Jersey Institute of Technology.
- 2017, 2019 — Communications Systems I (M.S.), New Jersey Institute of Technology.
- 2009 — Advanced Algebra (M.Sc.), Shahid Rajaee Teacher Training University, Tehran, Iran.

Research Mentoring

- 2026 — Group Summer Scholars Research Program (GSSRP), high-school student research mentoring, Kean University.
- 2026 — UR2PhD program mentoring, Kean University.
- 2026 — Graduate research assistant mentoring, Kean University.
- 2024–2026 — IUSE Scholars Academy student research mentoring, Kean University.
- 2025 — Group Summer Scholars Research Program (GSSRP), high-school student research mentoring, Kean University.
- 2025 — UR2PhD program mentoring, Kean University.
- 2023–Present — Master's thesis mentoring.
- 2022–Present — Undergraduate student research mentoring.

Professional Service

University and Community Service

- 2024–Present — Computational Science Ph.D. Curriculum Committee.
- 2023 — Judge, BRHS Hackathon.
- 2023–Present — Research Committee, WKU.

Editorial Boards

- 2025–Present — Editorial Board, *Mathematics and Computer Science*.
- 2019–Present — Editorial Board, *Electrical Engineering and Computer Science (EECS)*.

Professional Service (continued)

Peer Review

- Reviewer for *IEEE Transactions on Neural Networks and Learning Systems*; *Journal of Economic and Social Measurement*; *IEEE Access*; *IEEE Journal on Selected Areas in Information Theory*; *IEEE International Symposium on Information Theory*; *IEEE Transactions on Information Forensics and Security*; *IEEE Transactions on Communications*; *IEEE Transactions on Information Theory*; *Designs, Codes and Cryptography*; IEEE SPAWC; IEEE WCNC; CECNet; and ISCISC.

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE); IEEE Information Theory Society.

Technical Skills

Programming / Computing	Python, MATLAB, CoCoA, Magma.
Mathematical / Technical Tools	Mathematica, Maple, Vensim, LaTeX, Inkscape.
Languages	English, Persian, Azerbaijani, Spanish.